REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

sUPI PLATFORM BOARD

CONTENTS

TITLE	SHEET
BLOCK DIAGRAM	2
POWER INPUT	3
PAYLOAD POWER CONTROL	4
DATA SWITCHES	5

Block Diagram

File: 100046138_block_diagram.kicad_sch

RELATED PARTS			
TYPE	NUMBER	CFG	REV
PWB	100069026	001	1
PWA	100064667	001	1
EBOM	100064667	001	1

DRAWING MEETS REQUIREMENTS OF MIL-STD-31000

DO NOT SCALE THIS DRAWING

UNLESS OTHERWISE SPECIFIED
CAPACITORS ARE IN MICROFARADS
RESISTORS ARE IN OHMS
DIMENSIONS ARE IN INCHES

TOLERANCE ON
DECIMALS ANGLES
± .01 ± .1°
± .005

CONTRACT NO.

	BY DIRECTION	DATE
DWN		
CHK		
APVD		
APVD		
APVD		
APVD		

PCB SCHEMATIC
SUPI PLATFORM BOARD

SIZE	CAGE CODE	DRAWING NO.	REV
D		100046138	2
SCALE 1:1	FILENAME: 100064667.kicad_sch		
	SHEET:		SHEET: 1/5

DESIGN NOTES

1. CONNECTOR PINOUT IS MIRRORED FROM PAYLOAD

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
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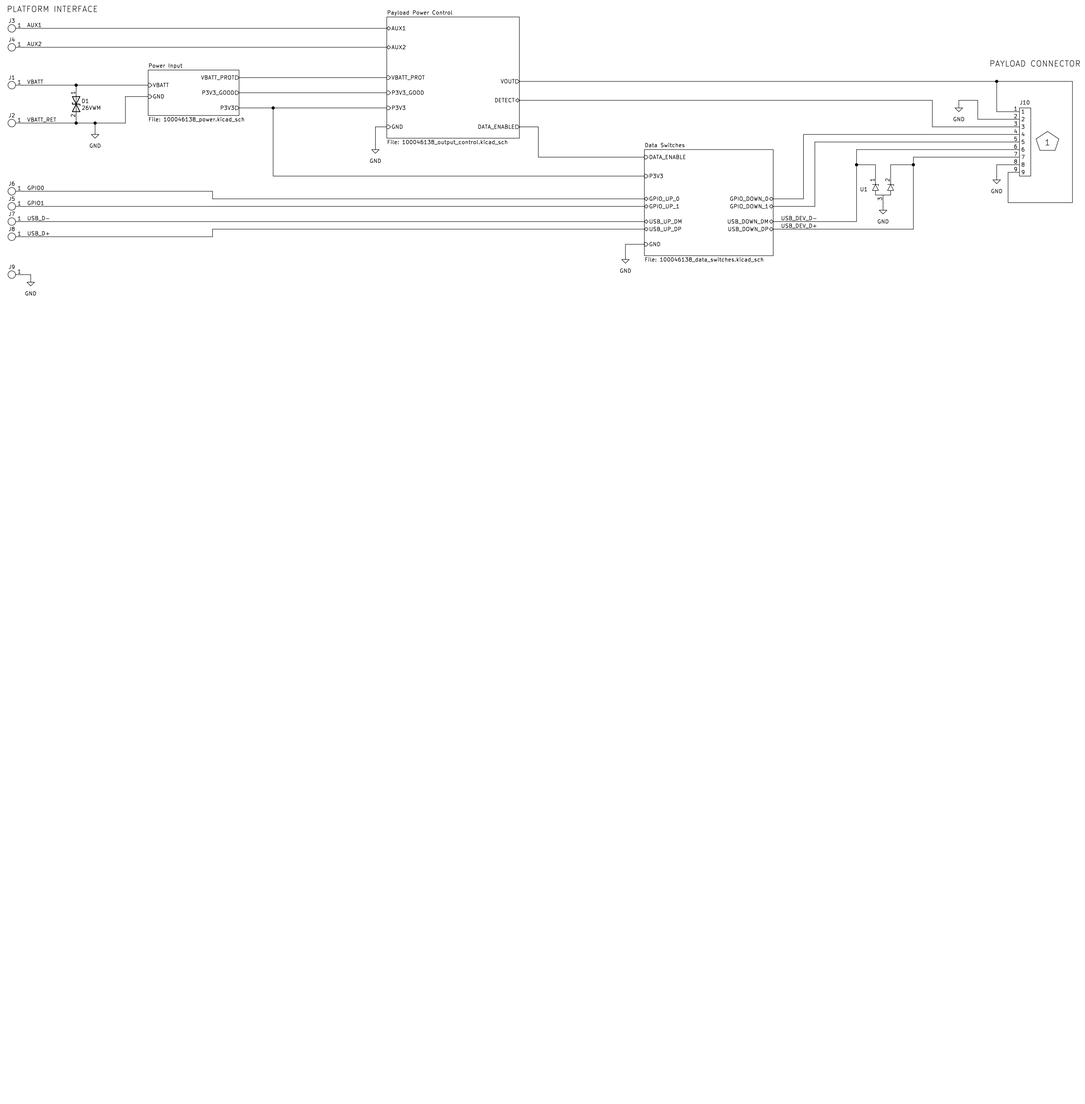
BLOCK DIAGRAM

FIDUCIALS

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
FID5 Fiducial
FID6 Fiducial

MOUNTING HOLES

MH1
MH2
MH3



SIZE D	CAGE CODE	DRAWING NO. 100046138	REV 2
SCALE 1:1	FILENAME: 100046138_block_diagram.kicad_sch		
	SHEET: Block Diagram		SHEET: 2/5

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

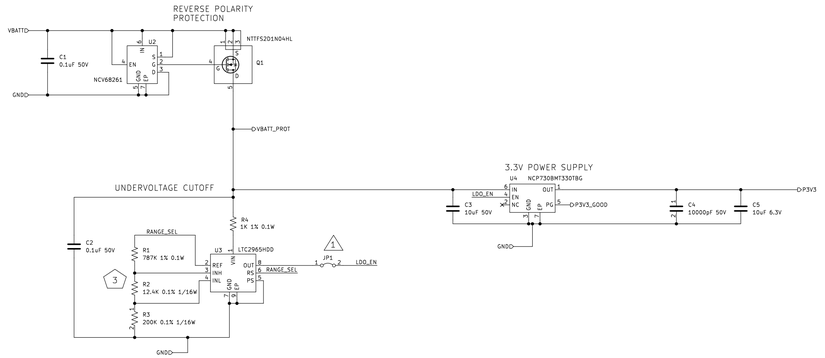
LAYOUT GUIDANCE

1. PLACE SINKSCREEN TEXT NEAR JP1: "LO BATT CUTOFF"

DESIGN NOTES

3. RISING VOLTAGE: 20.4V
FALLING VOLTAGE: 19.2V

POWER INPUT



SIZE: D	CAGE CODE:	DRAWING NO.: 100046138	REV: 2
SCALE: 1:1	FILENAME: 100046138_power.kicad_sch		
	SHEET: Power Input		SHEET: 3/5

PAYLOAD POWER CONTROL

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

LAYOUT GUIDANCE

PLACE JP2 AND JP3 TOGETHER.

2 PLACE SILKSCREEN NEAR JP2:
"DETECT PAYLOAD"

PLACE SILKSCREEN NEAR JP3:
"FORCE ON"

PLACE SILKSCREEN BETWEEN JP2 AND JP3:
"SELECT"

3 PLACE C26 AND D1 NEXT TO J11

4 PLACE U8 NEXT TO J4 AND J5

5 PLACE R19 AND R20 NEXT TO U4

DESIGN NOTES

4 AUX PINS COULD BE USART1 OR I2C1 FOR COMMUNICATION WITH THE HOST PLATFORM. SERIES RESISTORS ON AUX SIGNALS ARE INTENDED TO PROTECT MICROCONTROLLER IF PINS ARE SHORTED TO GND OR 3.3V AND TO PREVENT RINGING.

5 THE 1-WIRE BUS CAN EITHER BE IMPLEMENTED IN FIRMWARE USING TIM1_CH4 ON PA3 OR USART2 TX/RX ON PA2/PA3.

6 RESISTORS IN SERIES WITH THE JUMPERS ARE TO PREVENT DAMAGE TO MICROCONTROLLER IF BOTH JUMPERS ARE BRIDGED AT ONCE. ONLY ONE JUMPER IS MEANT TO BE BRIDGED AT A TIME.

7 RC ON ENABLE SERVES TWO PURPOSES:
1. EXTEND LOAD SWITCH OUTPUT DISCHARGE TIME. ACTIVE DISCHARGE ONLY OCCURS WHEN ENABLE LINE IS WITHIN 0.6-1.8V WINDOW.
2. PREVENT DAMAGING RINGING DUE TO LARGE INPUT CAPACITANCE

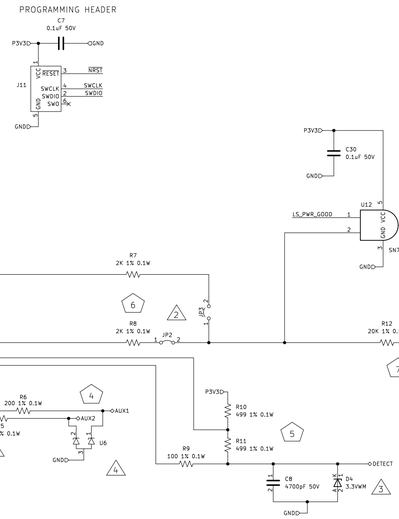
8 DATA SIGNAL (USB AND GPIO) ENABLE SEQUENCE:
1. ENABLE LINE ASSERTED VIA MICROCONTROLLER OR SOLDER BRIDGE
2. LOAD SWITCH ENERGIZES OUTPUT AFTER PROGRAMMED DELAY
3. POWER-ON RESET ADDS FURTHER DELAY. THEN ENABLES DATA SWITCHES

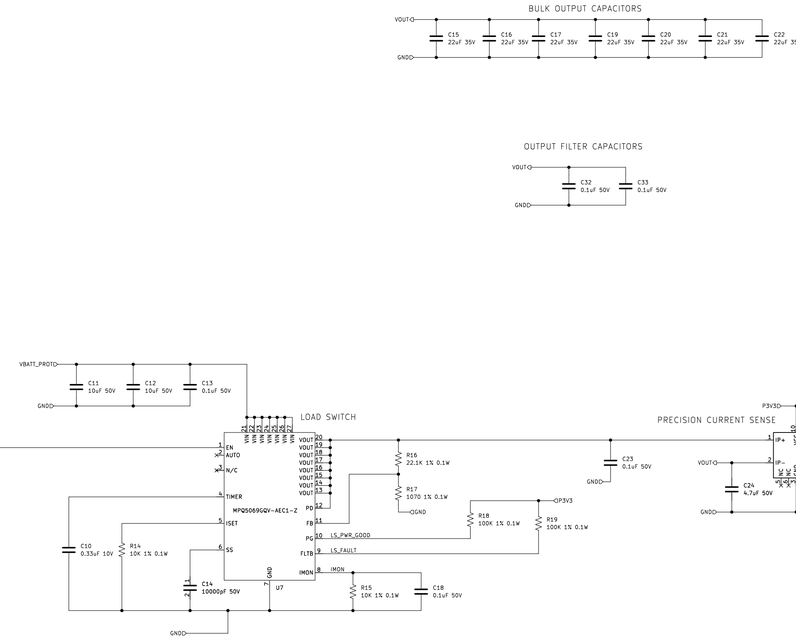
DATA SWITCHES ARE DISABLED IMMEDIATELY IF ANY OF THE FOLLOWING ARE TRUE:
1. ENABLE LINE IS DE-ASSERTED
2. LOAD SWITCH POWER GOOD IS DE-ASSERTED
3. 3.3V RAIL FALLS BELOW POWER-ON RESET THRESHOLD

PINS PC14, PC15, AND PB7 ARE RESERVED AS HARDWARE REVISION DETECTION PINS. THE FIRMWARE WILL USE THESE TO ENSURE THE CORRECT MICROCONTROLLER PIN ASSIGNMENTS.

REV	PC15	PC14	PB7
0	N/C	N/C	N/C
1	GND	N/C	N/C
2	3.3V	N/C	N/C
3	N/C	GND	N/C
4	GND	GND	N/C
5	3.3V	GND	N/C
6	N/C	3.3V	N/C
...	...	...	...
26	3.3V	3.3V	3.3V

HARDWARE REVISION: 1





SIZE	CAGE CODE	DRAWING NO.			REV
D		100046138			2
SCALE	FILENAME: 100046138_output_control.kicad_sch				
1:1	SHEET: Payload Power Control			SHEET: 4/5	

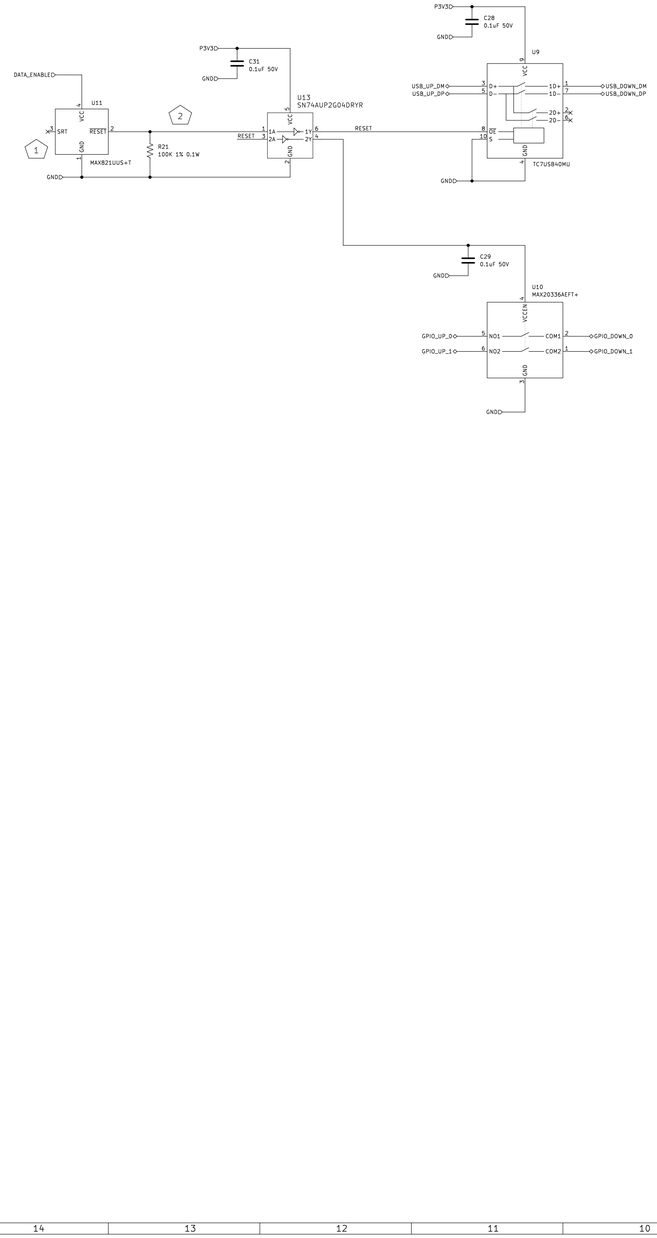
DESIGN NOTES

1. DELAY AFTER DATA_ENABLE ASSERT IS 100MS MINIMUM, 200MS MAXIMUM, 160MS TYPICAL. $\overline{\text{RESET}}$ WILL ASSERT IMMEDIATELY IF DATA_ENABLE FALLS BELOW THRESHOLD.

2. $\overline{\text{RESET}}$ IS CONSIDERED TO HAVE LOW DRIVE STRENGTH SO IT IS EFFECTIVELY BUFFERED FOR GPIO SWITCH BY PASSING THROUGH INVERTER TWICE.

REVISION HISTORY				
REV	DESCRIPTION	DWN	CHK	APVD
1	ENGR RELEASE PER			ENG APPROVAL PER

DATA SWITCHES



SIZE: D	CAGE CODE:	DRAWING NO.: 100046138	REV: 2
SCALE: 1:1	FILENAME: 100046138_data_switches.kicad_sch		
	SHEET: Data Switches		SHEET: 5/5